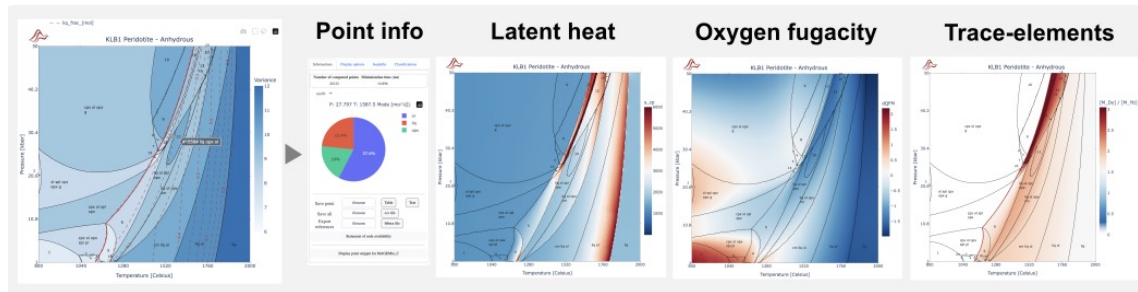
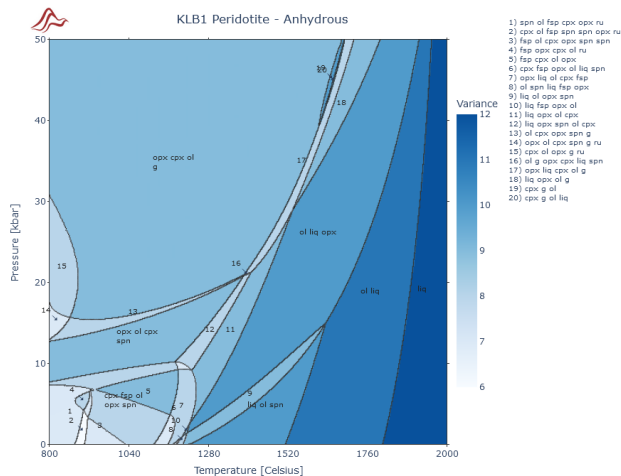


MAGEMinApp: introduction [v1.2.7]

Nicolas Riel - Johannes-Gutenberg University Mainz, Germany

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Main contributors

B. Kaus	E. Moulas
H. Dominguez	P. Lanari
J-F. Moyen	J. Forshaw
S. Schorn	T. Holland

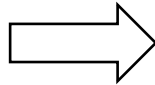
<https://github.com/ComputationalThermodynamics/Resources>

Thermodynamic equilibrium – stable phases

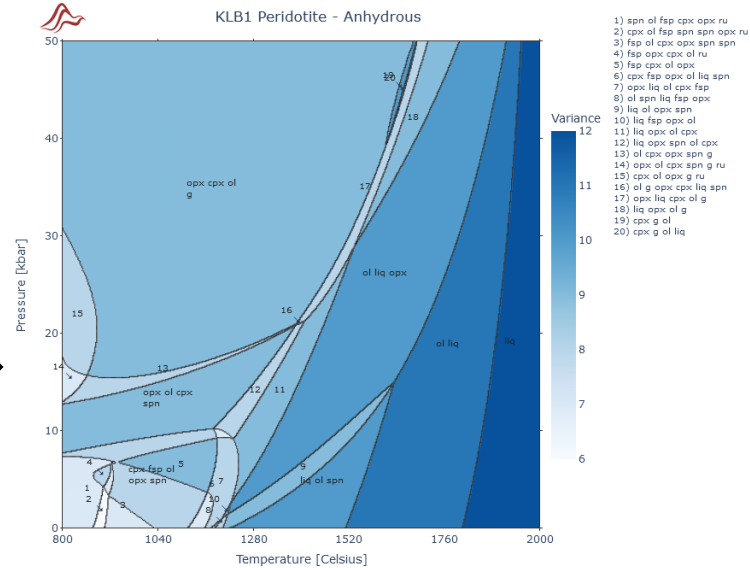
- Pressure
- Temperature
- Composition
- Thermodynamic database (minerals/melt)



Gibbs free energy minimization



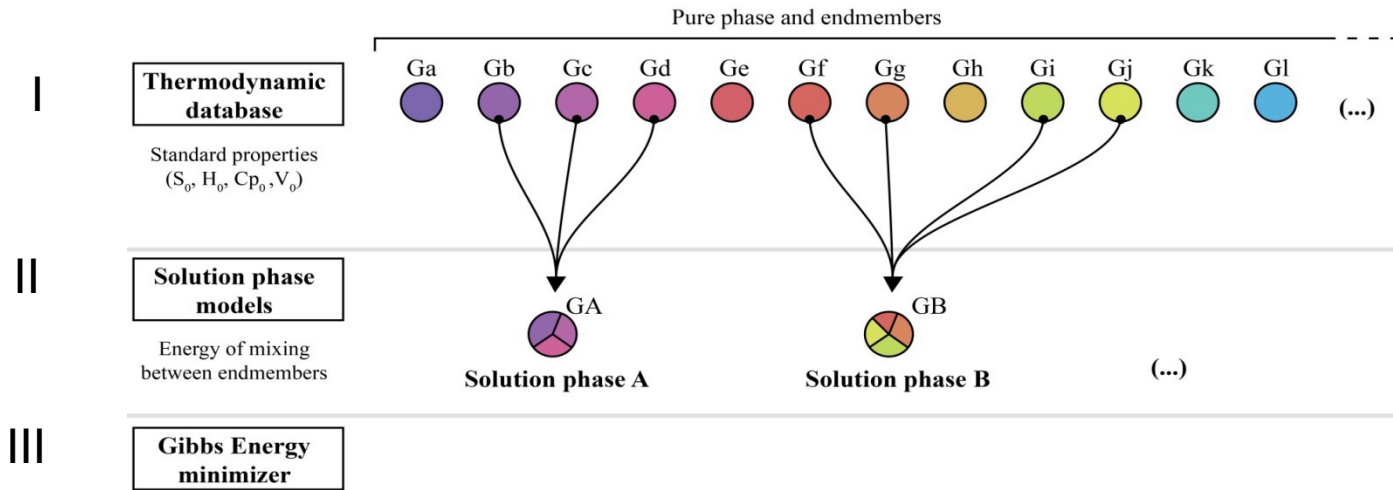
- Perple_X
- MELTS
- Theriak-Domino
- MAGEMin



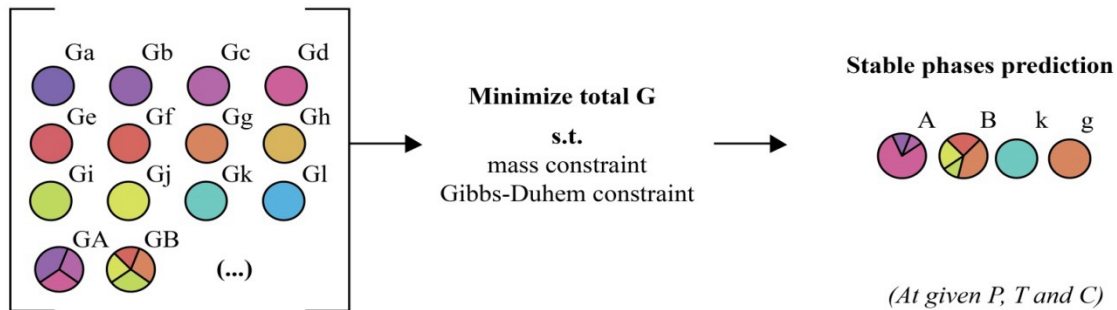
- Stable phases
- Phase fractions, compositions
- Solid/fluid/melt densities
- Seismic velocities, C_p , α ...

I. Principles of Gibbs Free energy minimization

Components for stable phase prediction



e.g.,
tc-ds6xx.txt for TC
hp02ver.dat for Perple_X



Gibbs free energy minimization: a challenging optimization problem

- Given a set of minerals at given P, T and C

Minimize
(at fixed P,T,C)

$$G_{\text{sys}} = \underbrace{\sum_{\lambda=1}^{\Lambda} \alpha_{\lambda} \sum_{i=1}^{N_{\lambda}} \mu_{i(\lambda)} p_{i(\lambda)}}_{\text{Solution phase (e.g., olivine)}} + \underbrace{\sum_{\omega=1}^{\Omega} \alpha_{\omega} \mu_{\omega}}_{\text{Pure phase (e.g., quartz)}}$$

where

$$\mu_{i(\lambda)} = g_{i(\lambda)}^0 + RT \log(\text{idm}_{i(\lambda)}) + g_{i(\lambda)}^{\text{ex}}$$

s.t. $\text{idm}_i > 0.0$

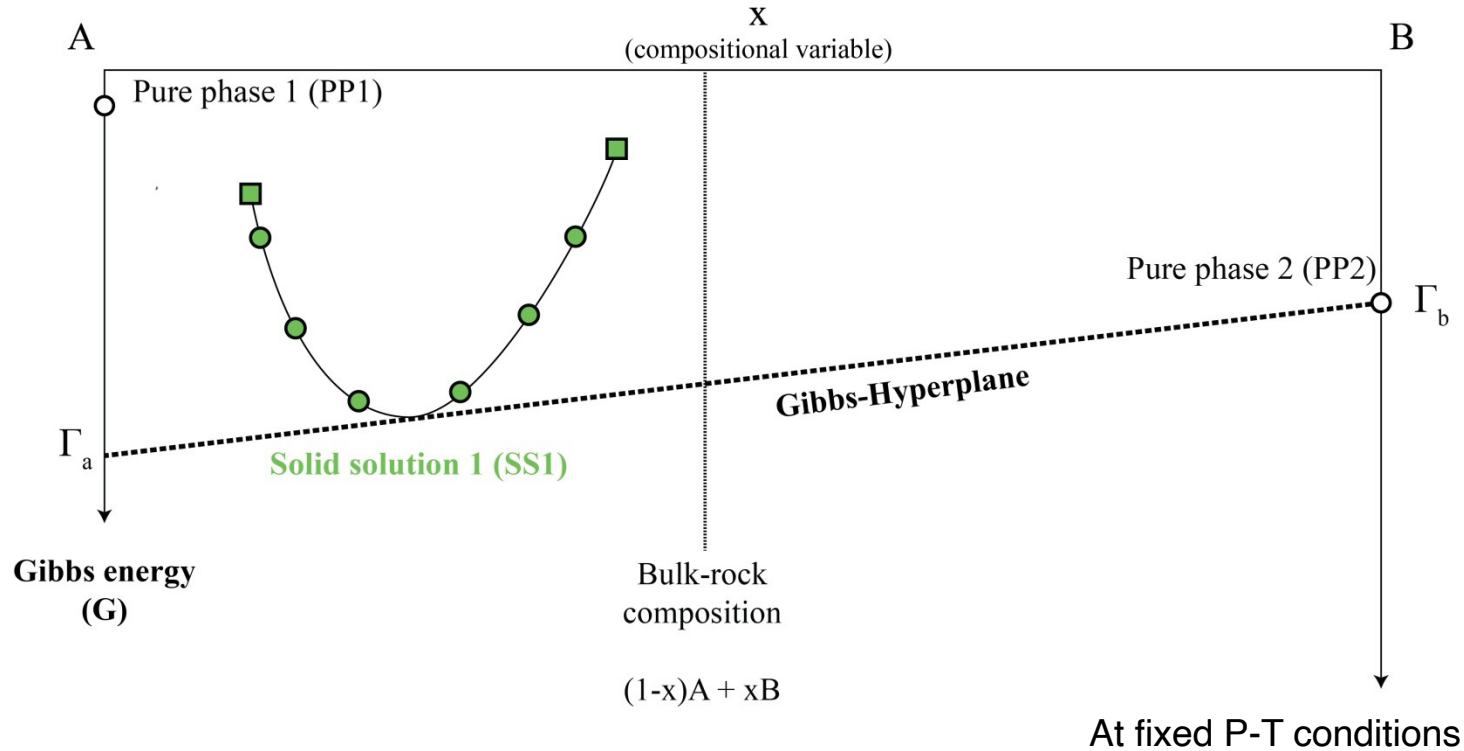
s.t.
Mass constraint

$$\sum_{\lambda=1}^{\Lambda} \alpha_{\lambda} \sum_{i=1}^{N_{\lambda}} a_{ij} p_{i(\lambda)} + \sum_{\omega=1}^{\Omega} \alpha_{\omega} a_{\omega j} - b_j = 0,$$

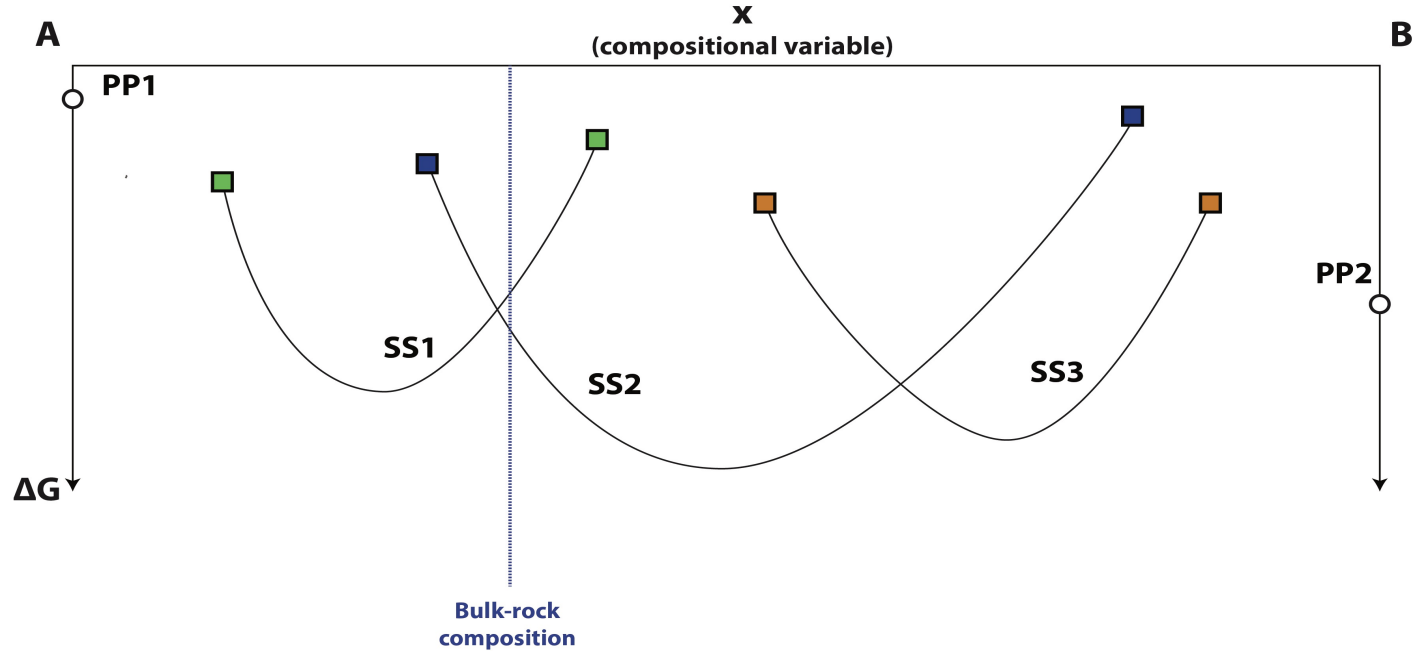
And Gibbs-Duhem
constraint

$$\mu_{i(\lambda),\omega} = \sum_{j=1}^C a_{i,\omega j} \Gamma_j,$$

Phase equilibria: how does that look like?

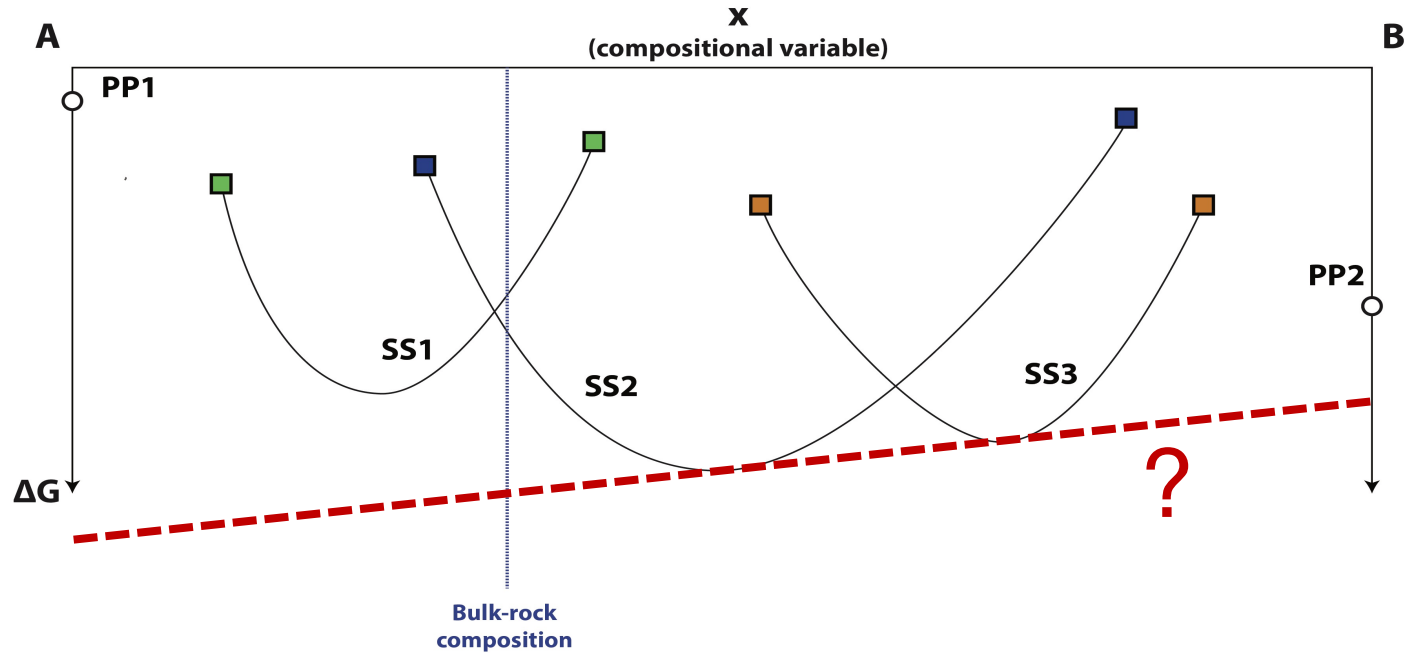


Phase equilibria



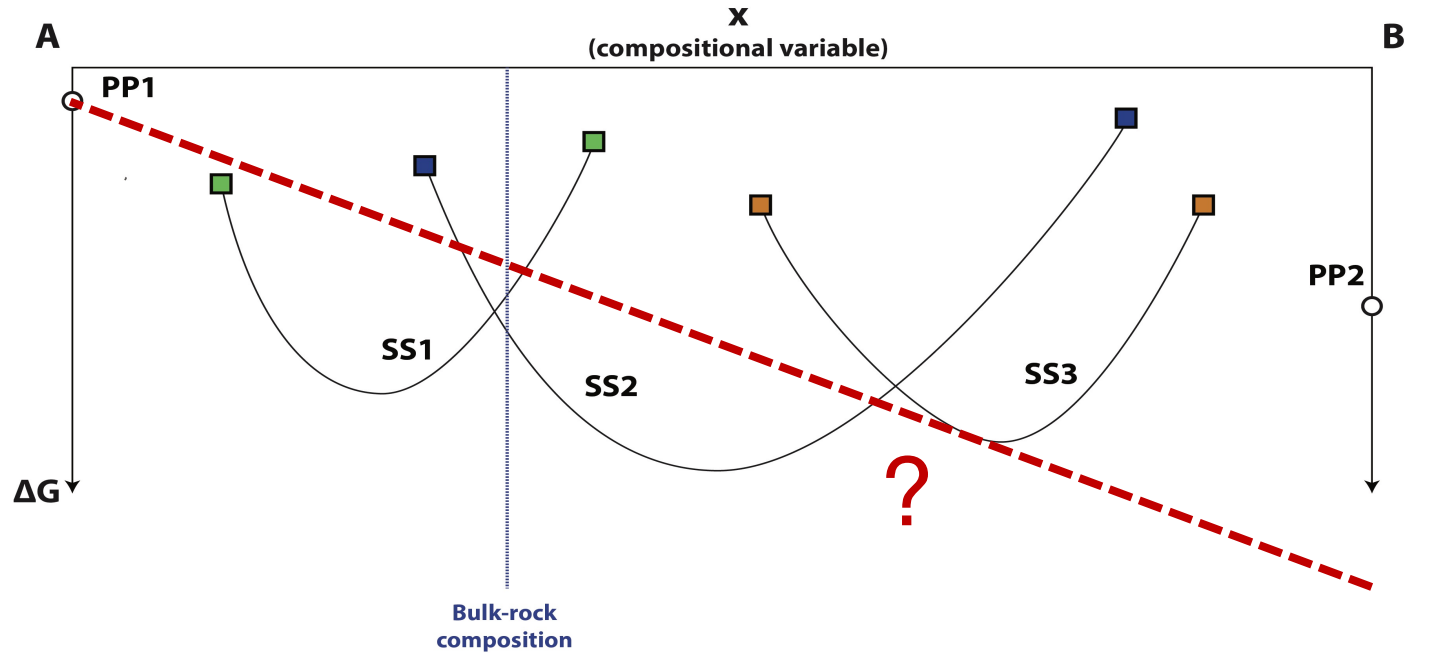
At fixed P-T conditions

Phase equilibria



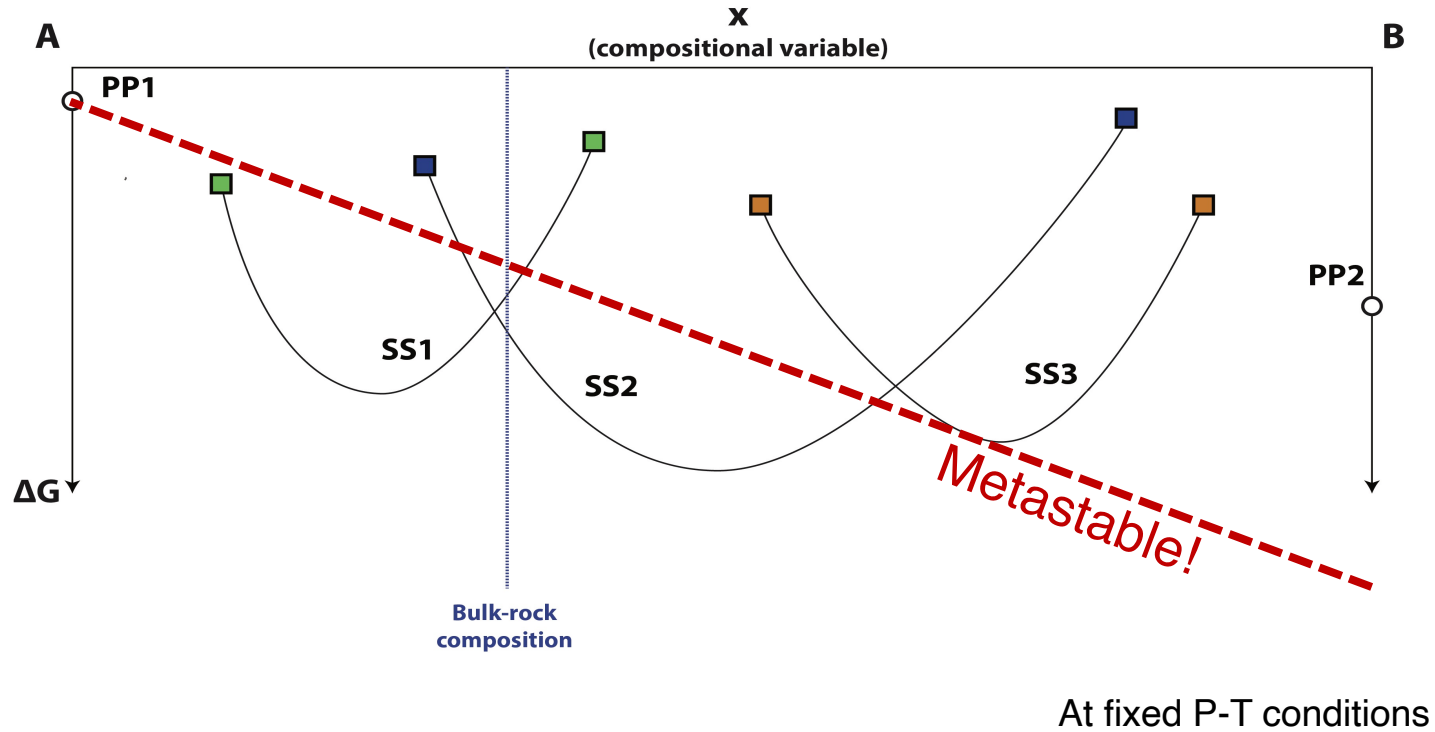
At fixed P-T conditions

Phase equilibria

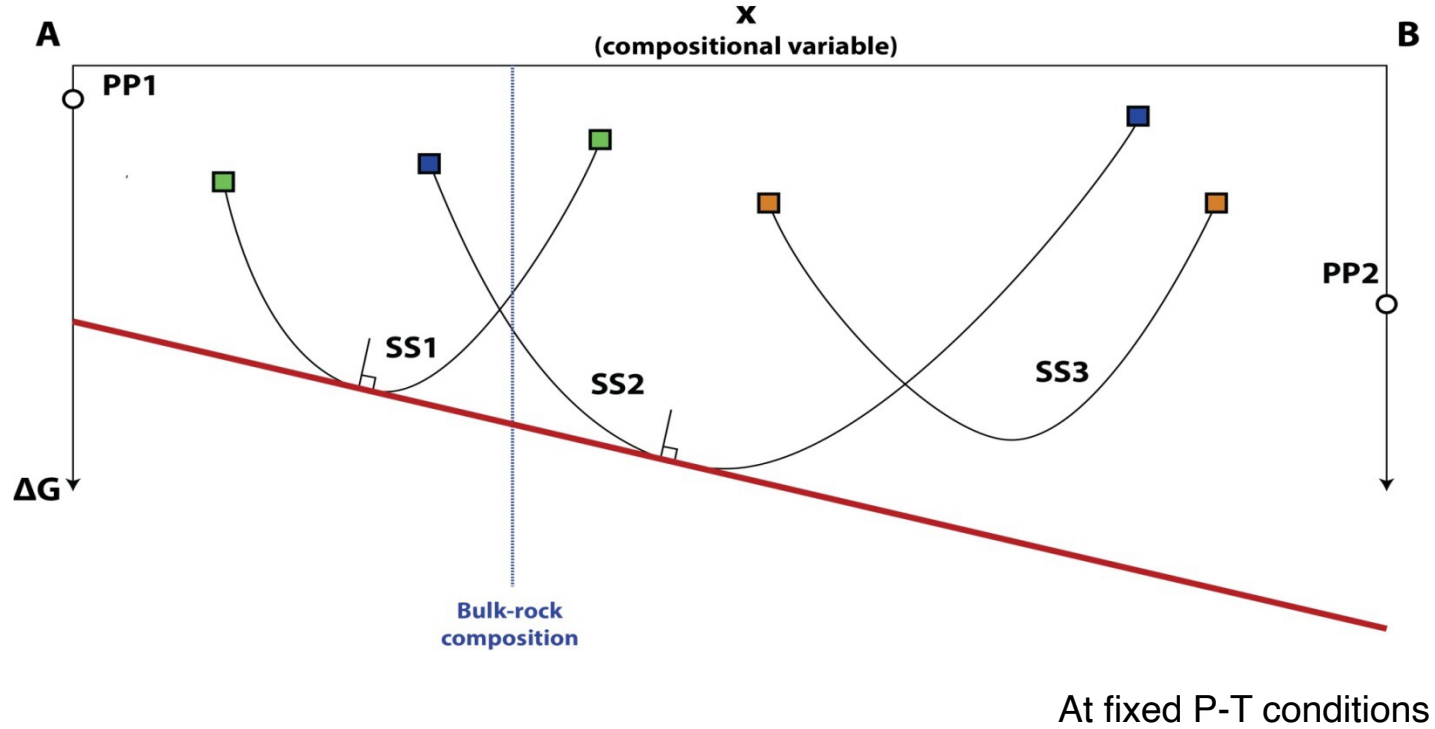


At fixed P - T conditions

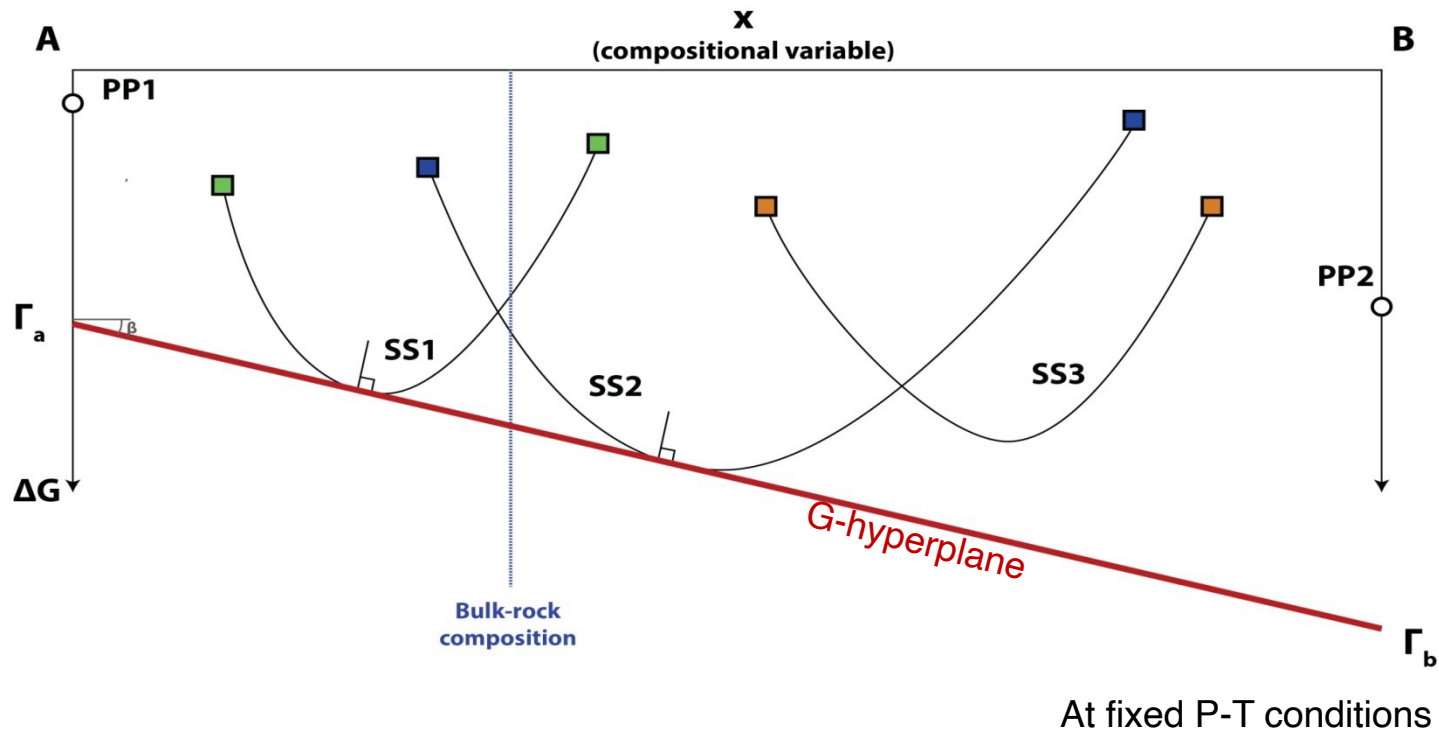
Phase equilibria



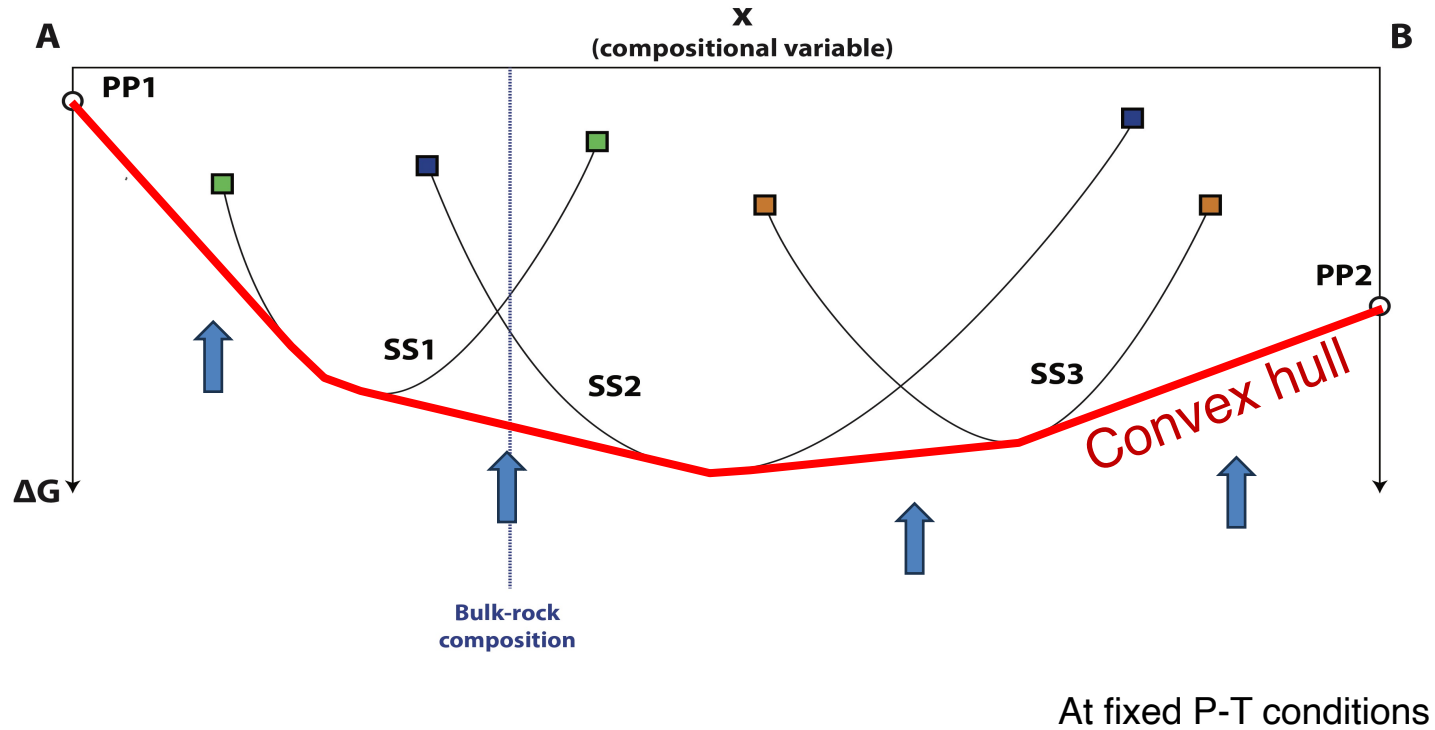
Phase equilibria



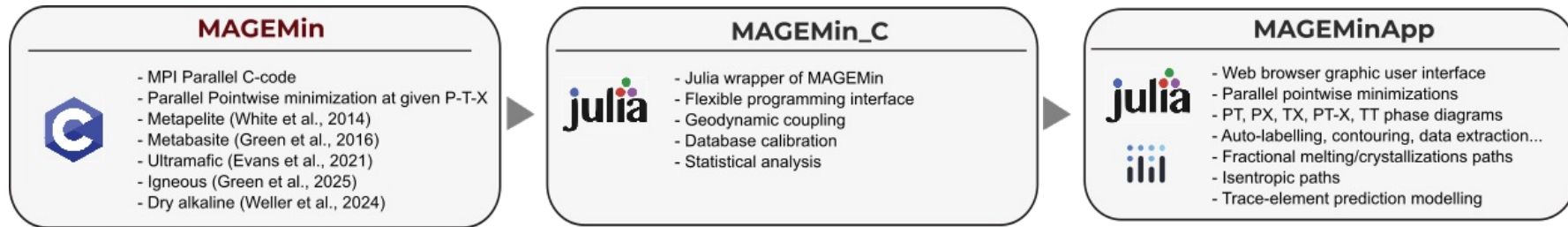
Phase equilibria



Phase equilibria

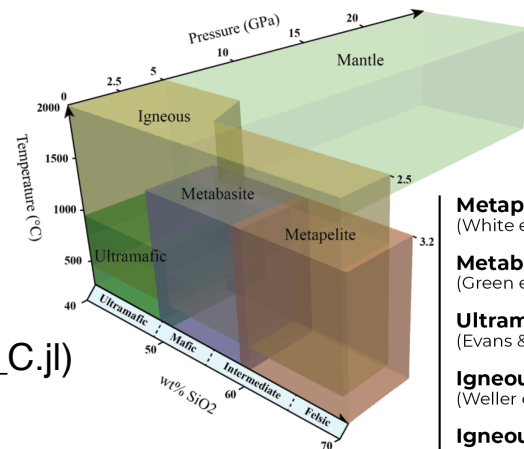


MAGEMin: current stage and ongoing developments



Current stage

- MPI parallel C-library with several built-in database
- Benchmarked 1 to 1 against THERMOCALC (for TC a-x sets)
- Julia wrapper for petrological/geodynamic coupling (MAGEMin_C.jl)
- Parallel web-browser Julia app (MAGEMinApp.jl)
- Single core performances
 - ~10 to 150 ms without initial guess (scales with #oxides and #solutions phases)
 - ~1 to 20 ms with initial guess



Metapelite *with melt!*
(White et al., 2014)

Metabasite *with melt!*
(Green et al., 2016)

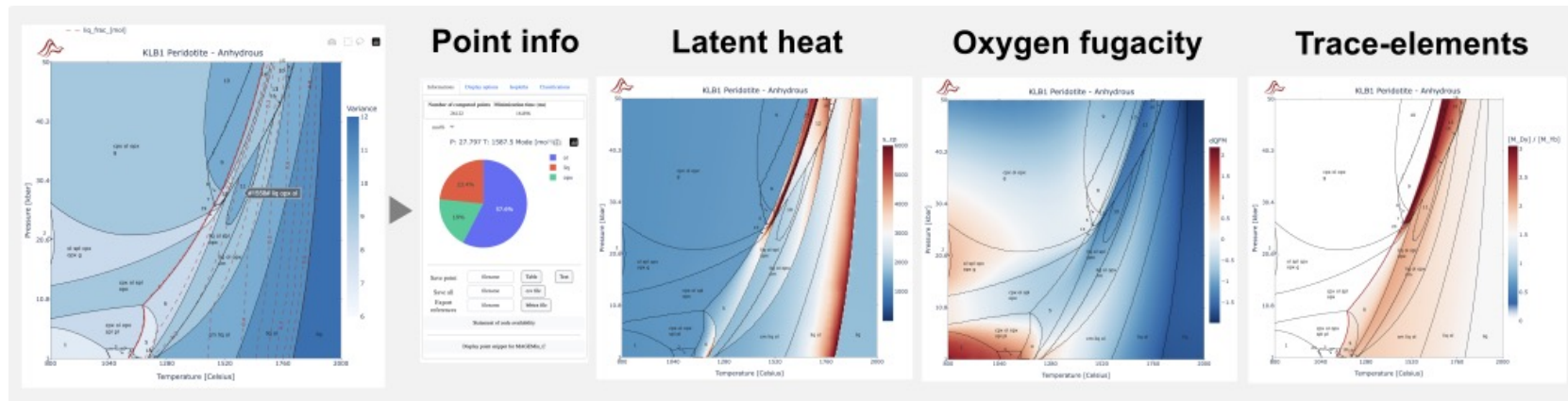
Ultramafic
(Evans & Forst, 2021)

Igneous alkali dry
(Weller et al., 2024)

Igneous *with melt!*
(Green et al., 2025 after H18)

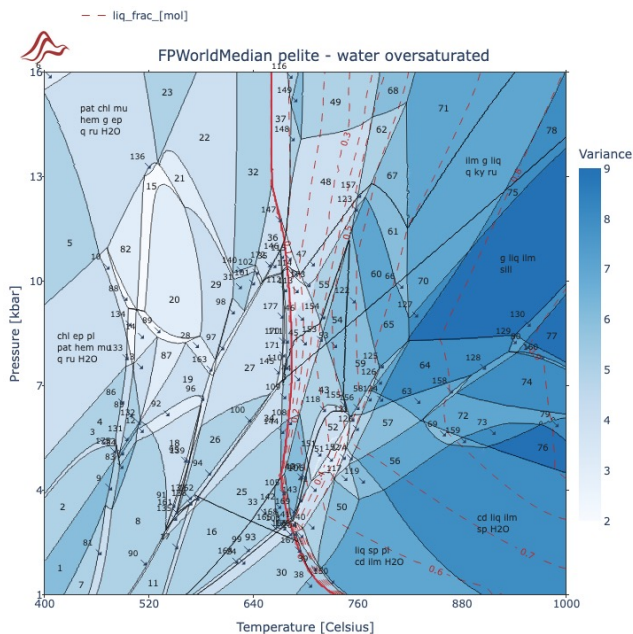
Mantle
(Holland et al., 2013)
(Stixrude & L.-Bertelloni, 2011)
(Stixrude & L.-Bertelloni, 2021)

MAGEMinApp.jl

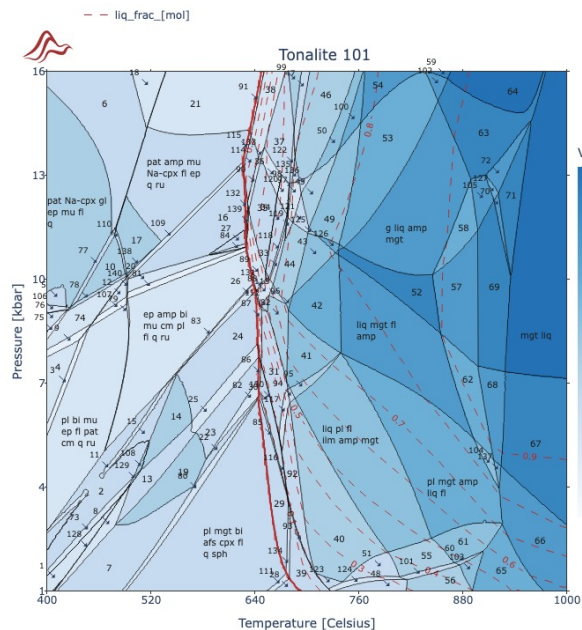


- P-T, T-X, P-X, PT-X, T-T phase diagrams
- Trace-element partitioning
- PTX paths (fractional melting/crystallization)
- Automatic labelling, isocontours, vector format figures, export data as `.csv` etc.

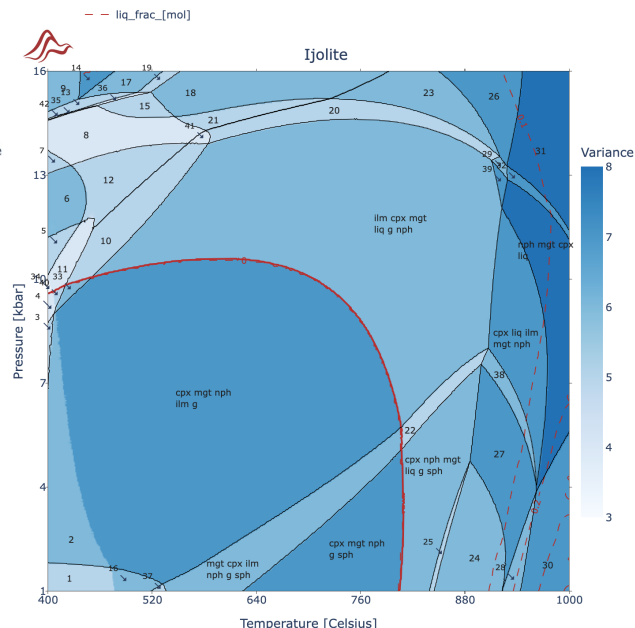
MAGEMinApp: examples of phase diagrams



- World Median metapelite
- Metapelite – White et al., 2014



- Tonalite 101
- Igneous – Green et al., 2025



- Ijolite
- Alkaline dry – Weller et al., 2024

MAGEMin_C.jl

- Programmable Julia interface to MAGEMin
- Parallel calculations (multi-threaded)
- Easy to use

```
using MAGEMin_C
data = Initialize_MAGEMin("ig", verbose=false);
P,T = 10.0, 1100.0
Xoxides = ["SiO2"; "Al2O3"; "CaO"; "MgO"; "FeO"; "Fe2O3"; "K2O"; "Na2O"; "TiO2"; "Cr2O3"; "H2O"];
X = [48.43; 15.19; 11.57; 10.13; 6.65; 1.64; 0.59; 1.87; 0.68; 0.0; 3.0];
sys_in = "wt"
out = single_point_minimization(P, T, data, X=X, Xoxides=Xoxides, sys_in=sys_in)
```

- Stores results in a structure

```
out.
G_system Gamma MAGEMin_ver M_sys PP_vec P_kbar SS_vec T_C V Vp Vp_S Vs Vs_S X
aAl2O3 aFeO aH2O aMgO aSiO2 aTiO2 alpha bulk bulkMod bulkModulus_M bulkModulus_S bulk_F bulk_F_wt bulk_M
bulk_M_wt bulk_S bulk_S_wt bulk_res_norm bulk_wt cp dQFM dataset enthalpy entropy fO2 frac_F frac_F_wt frac_M
frac_M_wt frac_S frac_S_wt iter mSS_vec n_PP n_SS n_mSS oxides ph ph_frac ph_frac_vol ph_frac_wt ph_id
ph_type rho rho_F rho_M rho_S s_cp shearMod shearModulus_S status time_ms
```

MAGEMin github: ComputationalThermodynamics

- First, access the online documentation:



https://computationalthermodynamics.github.io/MAGEMin_C.jl/dev/

MAGEMin Toolset
Phase equilibrium modelling

A set of tools to compute phase equilibria for Earth-like compositions

[Getting Started](#) [MAGEMin_C.jl API](#) [View on GitHub](#)

MAGEMinApp
Effortlessly compute phase diagrams and Pressure-Temperature paths

MAGEMin_C
Set of function to freely compute phase equilibrium in Julia

MAGEMin
C program backend of MAGEMin

What is MAGEMin?

Home

Method

MAGEMinApp.jl

- Introduction
- Installation
- Bulk-rock input file
- Phase diagrams tutorials
- P-T-X paths tutorials
- Citation
- Interface

MAGEMin_C.jl

- Introduction
- Installation
- Tutorials
- API

MAGEMin

- Introduction
- Compilation
- Tutorials

MAGEMinApp: launching the App in parallel

VISUAL STUDIO CODE / TERMINAL MAC, LINUX

```
seph42@DESKTOP-2V82075:~$ julia -t 6

Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.
Version 1.10.0 (2023-12-25)
Official https://julialang.org/ release

julia> using MAGEMinApp
App()
Using libMAGEMin.dylib from MAGEMin_jll

julia> App()
[ Info: Listening on: 127.0.0.1:8050, thread id: 2
```

```
julia -t 6
using MAGEMinApp
App()
```

POWERSHELL WINDOWS

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

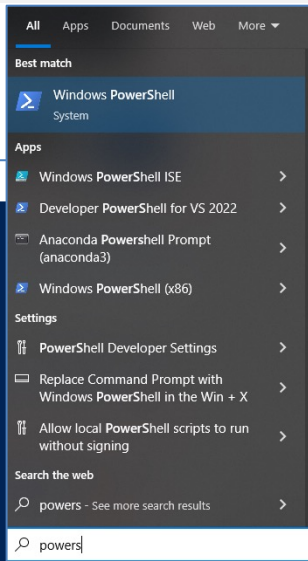
PS C:\Users\Seph> $env:JULIA_NUM_THREADS=6
PS C:\Users\Seph> julia

Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.
Version 1.9.3 (2023-08-24)
Official https://julialang.org/ release

julia> using MAGEMinApp
Using libMAGEMin.dylib from MAGEMin_jll

julia> App()
[ Info: Listening on: 127.0.0.1:8050, thread id: 1
```

```
$env:JULIA_NUM_THREADS=6
julia
using MAGEMinApp
App()
```



MAGEMinApp: interface (Phase diagrams)

Phase diagrams **PTX path** General information

Setup **Diagram** Trace-elements

Phase diagram parameters

Thermodynamic database

Igneous (Green et al., 2025, after H18) ▼

Dataset

ds636 ▼

Phase selection

Solution phase

Pure phase

- | | |
|--|--|
| ☒ spl | ☒ ne |
| ☒ bi | ☒ q |
| ☒ cd | ☒ crst |
| ☒ cpx | ☒ ird |
| ☒ ep | ☒ coe |
| ☒ g | ☒ stv |
| ☒ amp | ☒ ky |
| ☒ ilm | ☒ sill |
| ☒ liq | ☒ and |
| ☒ ol | ☒ ru |
| ☒ opx | ☒ sph |
| ☒ fsp | ☒ H2O |
| ☒ ft | |
| ☒ mu | |
| ☒ fper | |
| ☒ chl | |

Diagram type

P-T diagram (fixed composition) ▼

Solidus H₂O-saturated

false ▼

TE predictive model

false ▼

Pressure [kbar]

min max
0.01 20.01

Temperature [°C]

800 1400

Bulk-rock composition

mol% ▼

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous ▼

oxide	fraction
SiO2	38.494
Al2O3	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K2O	0.01
Na2O	0.25
TiO2	0.1
O	0.096
Cr2O3	0.109
H2O	0

ig

General parameters

Title

KLB1 Peridotite - Anhydrous

Update

Reset

Compute phase diagram

Save/Load Diagram

filename

Save state

Load state

State/CSV directory: /Users/nicolasriel/julia/packages/MAGEMinApp/rkeff

Help and contact

Links

[Tutorials](#)
[Post issue](#)
[Open discussion](#)
[Discord](#)

Comments

Set of examples on how to use MAGEMinApp
Something is not working properly?
Need additional options?
... or join our Discord!

Contributors

MAGEMinApp: interface (Phase diagrams)

Phase diagrams

PTX path

General information

Setup

Diagram

Trace-elements

Phase diagram parameters

Thermodynamic database

Igneous (Green et al., 2025, after H18)

Dataset

ds636

Phase selection

Solution phase

☒ spl
☒ bi
☒ cd
☒ cpx
☒ op
☒ g
☒ amp
☒ ilm
☒ liq
☒ ol
☒ opx
☒ fsp
☒ il
☒ mu
☒ fper
☒ chl

Pure phase

☒ ne
☒ q
☒ crst
☒ trd
☒ coe
☒ stv
☒ ky
☒ sill
☒ and
☒ ru
☒ sph
☒ H2O

Diagram type

P-T diagram (fixed composition)

Solidus H₂O-saturated

false

TE predictive model

false

min

max

Pressure [kbar]

0.01

20.01

Temperature [°C]

800

1400

Bulk-rock composition

mol%

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous

oxide	fraction
SiO2	38.494
Al2O3	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K2O	0.01
Na2O	0.25
TiO2	0.1
O	0.096
Cr2O3	0.109
H2O	0

ig

General parameters

Title

KLB1 Peridotite - Anhydrous

Update

Reset

Compute phase diagram

Save/Load Diagram

filename

Save state

Load state

State/CSV directory: /Users/nicolasriell/julia/packages/MAGEMinApp/rkeIF

Help and contact

Links

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[Post issue](#)
[Open discussion](#)
[Discord](#)

Comments

Set of examples on how to use MAGEMinApp
Something is not working properly?
Need additional options?
... or join our Discord!

Contributors

Grid properties

Bulk rock composition(s)

- Save/load state
- Compute diagram

21

MAGEMinApp: Thermodynamic database

Phase diagram parameters

Thermodynamic database

Dataset

Phase selection

Diagram type

Solidus H₂O-saturated

TE predictive model

Pressure [kbar]

Temperature [°C]

Initial grid subdivision

Refinement type

Refinement levels

Boost mode

Buffer

Solver

Verbose

Specific Cp

min max

0.01 20.01

800 1400

4 16 × 16 grid

Phases only

3

true

no buffer

Legacy

none

G0 (no latent heat)

Ter

id=360ac379-0079-480a-82f9-50835dd4cd1a; MAGEMinApp GUI v=1.2.7; using 12/12 threads

- **Mantle (Holland et al., 2013)**
NCFMAS
- **Metapelite (White et al., 2014)**
MnNCKFMASHTO
- **Metabasite (Green et al., 2016)**
NCKFMASHTO
- **Igneous (Green et al., 2025 after Holland et al., 2018)**
NCKFMASHTOCr
- **Ultramafic (Evans & Frost, 2021)**
FMASOSH
- **Dry igneous alkaline (Weller et al., in review)**
NCKFMASHTOCr
- **Mantle (Stixrude and Lithgow-Bertelloni, 2011, 2021)**
NCFMAS

MAGEMinApp: solution phase selection

Thermodynamic database

Dataset

Phase selection

Solution phase

Pure phase

Diagram type

Solidus H₂O-saturated

TE predictive model

Pressure [kbar]

Temperature [°C]

Igneous (Green et al., 2025, after H18)

ds636

spl
bi
cd
cpx
ep
g
amp
ilm
liq
ol
opx
fsp
fl
mu
fper

ne
q
crst
trd
coe
stv
ky
sill
and
ru
sph
H2O

P-T diagram (fixed composition)

false

false

min max

0.01 20.01

min max

800 1400

- Allow to deactivate phase
- Note that solution phase models from different dataset cannot be mixed (yet)

MAGEMinApp: diagram types

Phase diagram parameters

Thermodynamic database	Igneous (Green et al., 2025, after H18)	
Dataset	ds636	
Phase selection	Solution phasePure phase	
Diagram type	P-T diagram (fixed composition)	
Solidus H ₂ O-saturated	false	

TE predictive model	false	
---------------------	-------	--

Pressure [kbar]	min	max
	0.01	20.01
Temperature [°C]	min	max
	400	1000

Initial grid subdivision	4	16 × 16 grid
Refinement type	Phases only	
Refinement levels	4	
Boost mode	true	

Buffer	no buffer	
Solver	Legacy	
Verbose	none	
Specific Cp	G0	

P-T

Pressure [kbar]

Temperature [°C]

P-X

Fixed temperature

Pressure [kbar]

Composition [bulk1-bulk2]

T-X

Fixed pressure

Temperature [°C]

Composition [bulk1-bulk2]

PT-X

PT path

Composition [bulk1-bulk2]

MAGEMinApp: diagram types

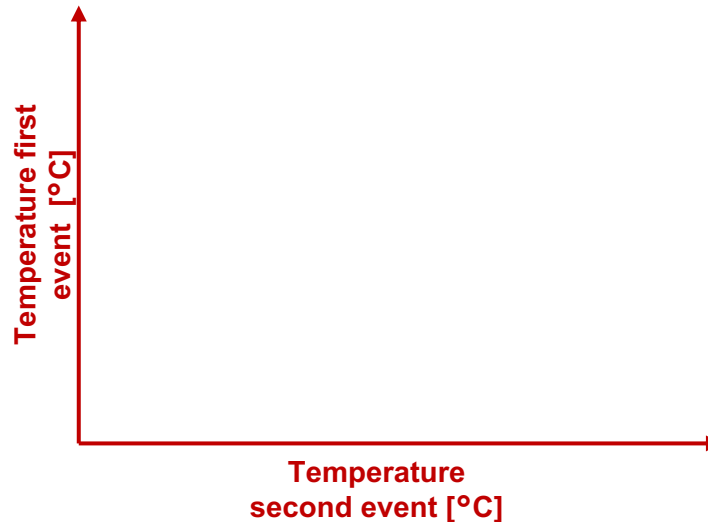
Phase diagram parameters

Thermodynamic database	Igneous (Green et al., 2025, after H18) ▼	
Dataset	ds636 ▼	
Phase selection	☐ Solution phase ☐ Pure phase	
Diagram type	P-T diagram (fixed composition) ▼	
Solidus H ₂ O-saturated	false ▼	
TE predictive model ▼		
false ▼		
min max		
Pressure [kbar]	0.01 ▼	20.01 ▼
min max		
Temperature [°C]	400 ▼	1000 ▼
Initial grid subdivision		
	4 ▼	16 × 16 grid
Refinement type	Phases only ▼	
Refinement levels	4 ▼	
Boost mode	true ▼	
Buffer		
	no buffer ▼	
Solver		
	Legacy ▼	
Verbose		
	none ▼	
Specific Cp		
	G0 ▼	

T-T

Poly-metamorphic diagram

**Fixed
pressure**



MAGEMinApp: Trace element predictive model

Implemented with the help of J-F Moyen (LMV) and H. Dominguez (Uni - Bern)

Phase diagram parameters

Thermodynamic database: Igneous (Green et al., 2025, after H18) ▼

Dataset: ds636 ▼

Phase selection: ☐ Solution phase ☐ Pure phase

Diagram type: P-T diagram (fixed composition) ▼

Solidus H₂O-saturated: false ▼

TE predictive model: false ▼

Pressure [kbar]: min 0.01 max 20.01

Temperature [°C]: min 400 max 1000

Initial grid subdivision: 4 16 × 16 grid

Refinement type: Phases only ▼

Refinement levels: 4 ▼

Boost mode: true ▼

Buffer: no buffer ▼

Solver: Legacy ▼

Verbose: none ▼

Specific Cp: G0 ▼

- Activate trace-element partitioning at supra-solidus conditions
- Displays trace element bulk panel on the right
- Gives the option to select among saturation models

TE predictive model: true ▼

Upload Kd's:

Kd's database: O. Laurent (2012) ▼

Zr saturation: none ▼

P2O5 saturation: none ▼

S saturation: none ▼

General parameters

Title: KLB1 Peridotite - Anhydrous

State/CSV directory: /home/seph42/MAGEMinApp/ij_v04.0

Trace-element composition

Drag and drop or select trace-element file

Pyrolite (Primitive Mantle) ▼

elements	µg/g
Rb	0.6
Ba	6.6
Th	0.0795
U	0.0203
Nb	0.658
Ta	0.037
La	0.648
Ce	1.675
Pb	0.15
Pr	0.254
Sr	19.9
Nd	1.25

MAGEMinApp: grid subdivision and refinement

Phase diagram parameters

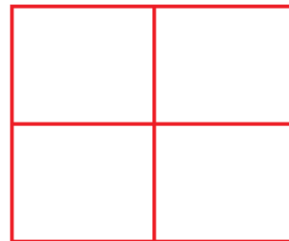
Thermodynamic database	Igneous (Green et al., 2025, after H18) ▼
Dataset	ds636 ▼
Phase selection	☐ Solution phase ☐ Pure phase
Diagram type	P-T diagram (fixed composition) ▼
Solidus H ₂ O-saturated	false ▼

TE predictive model	false ▼
---------------------	---------

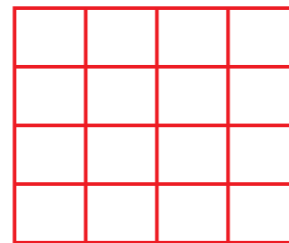
	min	max
Pressure [kbar]	0.01 ▼	20.01 ▼
	min	max
Temperature [°C]	400 ▼	1000 ▼

Initial grid subdivision	4 ▼	16 × 16 grid
Refinement type	Phases only ▼	
Refinement levels	4 ▼	
Boost mode	true ▼	

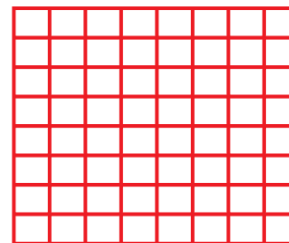
Buffer	no buffer ▼
Solver	Legacy ▼
Verbose	none ▼
Specific Cp	G0 ▼



1 initial grid subdivisions



2 initial grid subdivisions



3 initial grid subdivisions

**Number of cells
 $(2^{n_subdivision}-1)^2$**

MAGEMinApp: grid subdivision and refinement

Phase diagram parameters

Thermodynamic database	Igneous (Green et al., 2025, after H18)
Dataset	ds636
Phase selection	☐ Solution phase ☐ Pure phase
Diagram type	P-T diagram (fixed composition)
Solidus H ₂ O-saturated	false

TE predictive model	false
---------------------	-------

	min	max
Pressure [kbar]	0.01	20.01
	min	max
Temperature [°C]	400	1000

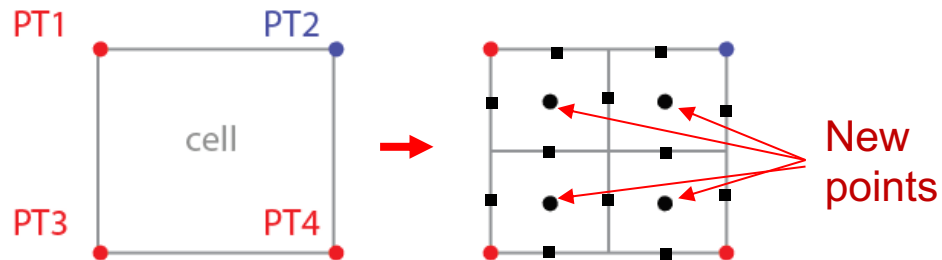
Initial grid subdivision	4	16 x 16 grid
--------------------------	---	--------------

Refinement type	Phases only
Refinement levels	4

Boost mode	true
------------	------

Buffer	no buffer
Solver	Legacy
Verbose	none
Specific Cp	G0

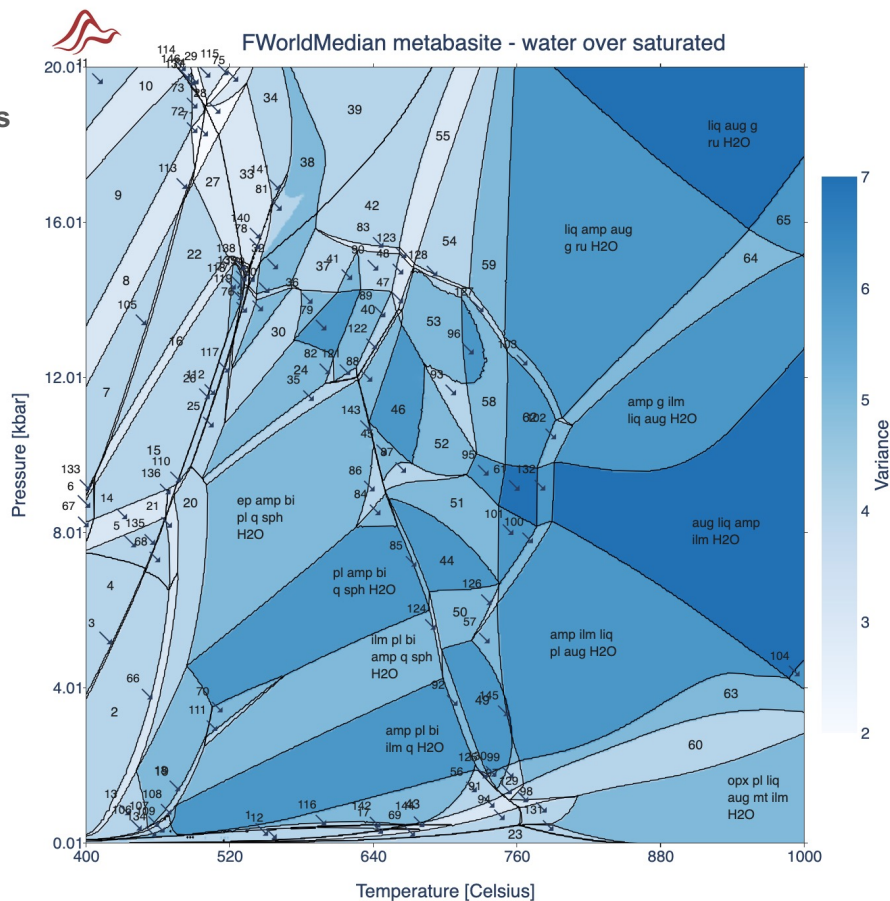
- **Phase only**
refines (split) cells when a phase change is recorded in one of the four corners
- **Dominant end-member**
refines (split) cells when the dominant end-member of any solution phase model changes in one of the four corners



- **Refinement levels**
Number of successive grid refinements

MAGEMinApp: grid refinement

Ca. 50,000 points



Export figure

Export all layers

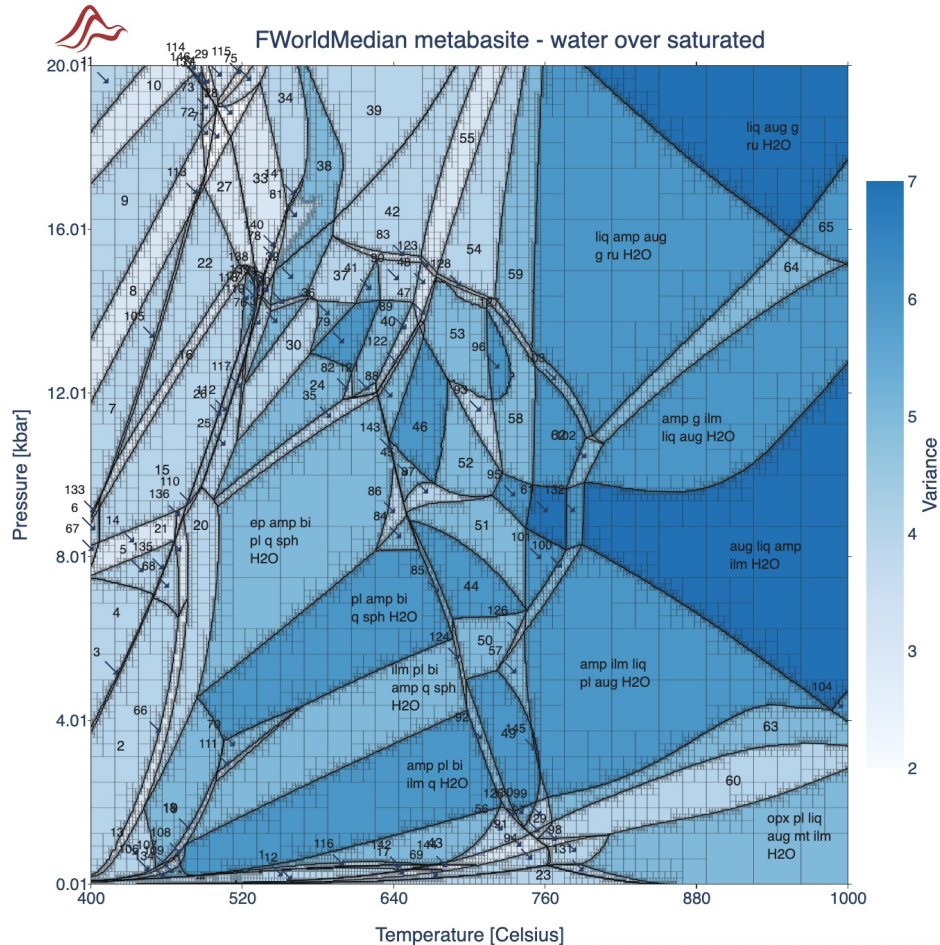
Figure directory: /Users/nicolasriell/julia/packages/
MAGEMinApp/rkeff/output/

Phase assemblages



- 1) bi ilm act pl act mt q H2O
- 2) act bi pl ep chl q sph H2O
- 3) chl ep abc act pl bi q sph H2O
- 4) ep abc chl bi act q sph H2O
- 5) ep tr bi chl abc q sph H2O
- 6) act chl bi gl ep q sph H2O
- 7) chl ep gl mu aug q sph H2O
- 8) aug mu ep gl chl q law sph H2O
- 9) aug gl chl mu q law sph H2O
- 10) chl aug gl mu tr q law sph H2O
- 11) tr gl chl mu q law sph H2O
- 12) opx pl bi act mt ilm q H2O
- 13) bi pl act chl amp q sph H2O
- 14) ep gl bi chl tr abc q sph H2O
- 15) ep chl bi tr gl q sph H2O
- 16) gl tr ep chl mu bi q sph H2O
- 17) mt pl ilm bi act opx aug q H2O
- 18) ilm chl amp bi pl q sph H2O
- 19) amp pl bi chl q sph H2O
- 20) amp pl chl bi ep q sph H2O
- 21) pl bi ep amp chl tr q sph H2O
- 22) gl ep chl tr mu q sph H2O
- 23) aug ilm pl opx afs mt q H2O
- 24) amp bi mu ep q sph H2O
- 25) chl bi ep amp mu q sph H2O
- 26) amp gl chl ep bi q sph H2O
- 27) ep gl g tr mu chl q sph H2O
- 28) ep tr mu g gl chl q law ru H2O
- 29) aug gl g mu tr q law ru H2O
- 30) ep amp mu bi q ru sph H2O
- 31) ep amp bi mu q ru H2O
- 32) amp g mu ep q ru H2O
- 33) g ep chl mu tr gl q ru H2O
- 34) g mu gl ep tr q ru H2O
- 35) pl amp ep bi mu q sph H2O

MAGEMinApp: grid refinement



Export figure

Export all layers

Figure directory: /Users/nicolasriell/julia/packages/
MAGEMinApp/rkeff/output/

Phase assemblages



- 1) bi ilm act pl act mt q H2O
- 2) act bi pl ep chl q sph H2O
- 3) chl ep abc act pl bi q sph H2O
- 4) ep abc chl bi act q sph H2O
- 5) ep tr bi chl abc q sph H2O
- 6) act chl bi gl ep q sph H2O
- 7) chl ep gl mu aug q sph H2O
- 8) aug mu ep gl chl q law sph H2O
- 9) aug gl chl mu q law sph H2O
- 10) chl aug gl mu tr q law sph H2O
- 11) tr gl chl mu q law sph H2O
- 12) opx pl bi act mt ilm q H2O
- 13) bi pl act chl amp q sph H2O
- 14) ep gl bi chl tr abc q sph H2O
- 15) ep chl bi tr gl q sph H2O
- 16) gl tr ep chl mu bi q sph H2O
- 17) mt pl ilm bi act opx aug q H2O
- 18) ilm chl amp bi pl q sph H2O
- 19) amp pl bi chl q sph H2O
- 20) amp pl chl bi ep q sph H2O
- 21) pl bi ep amp chl tr q sph H2O
- 22) gl ep chl tr mu q sph H2O
- 23) aug ilm pl opx afs mt q H2O
- 24) amp bi mu ep q sph H2O
- 25) chl bi ep amp mu q sph H2O
- 26) amp gl chl ep bi q sph H2O
- 27) ep gl g tr mu chl q sph H2O
- 28) ep tr mu g gl chl q law ru H2O
- 29) aug gl g mu tr q law ru H2O
- 30) ep amp mu bi q ru sph H2O
- 31) ep amp bi mu q ru H2O
- 32) amp g mu ep q ru H2O
- 33) g ep chl mu tr gl q ru H2O
- 34) g mu gl ep tr q ru H2O

MAGEMinApp: buffers

Phase diagram parameters

Thermodynamic database: Igneous (Green et al., 2025, after H18)

Dataset: ds636

Phase selection: ☐ Solution phase ☐ Pure phase

Diagram type: P-T diagram (fixed composition)

Solidus H₂O-saturated: false

TE predictive model: false

Pressure [kbar]: min 0.01 max 20.01

Temperature [°C]: min 400 max 1000

Initial grid subdivision: 4 16 × 16 grid

Refinement type: Phases only

Refinement levels: 4

Boost mode: true

Buffer: no buffer

Solver: Legacy

Verbose: none

Specific Cp: G0

- **Oxygen buffers**
QFM, MW, QIF, CCO, HM, NNO, IW
- **Fix system activity**
H₂O, O₂, SiO₂, FeO, MgO, Al₂O₃, TiO₂

Bulk-rock composition

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous

oxide	mol_fraction
SiO ₂	38.494
Al ₂ O ₃	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K ₂ O	0.01
Na ₂ O	0.25
TiO ₂	0.1
O	0.096
Cr ₂ O ₃	0.109
H ₂ O	0

Buffer multiplier: 0

- **Selecting a buffer displays the option to apply a shift in the RTlog scale**
e.g., +2 QFM
- **Allow to provide the activity value in case activity is selected**

MAGEMinApp: Specific heat capacity (latent heat of reaction)

Phase diagram parameters

Thermodynamic database	Igneous (Green et al., 2025, after H18) ▼	
Dataset	ds636 ▼	
Phase selection	Solution phase ☐ Pure phase ☐	
Diagram type	P-T diagram (fixed composition) ▼	
Solidus H ₂ O-saturated	false ▼	
TE predictive model	false ▼	
Pressure [kbar]	min 0.01 ▼	max 20.01 ▼
Temperature [°C]	min 400 ▼	max 1000 ▼
Initial grid subdivision	4 ▼	16 × 16 grid
Refinement type	Phases only ▼	
Refinement levels	4 ▼	
Boost mode	true ▼	
Buffer	no buffer ▼	
Solver	Legacy ▼	
Verbose	none ▼	
Specific Cp	G0 ▼	

- Method to compute specific heat capacity

$$C_p = -T \frac{\Delta^2 G}{\Delta T^2} = -T \frac{G^{(T+\delta T)} + G^{(T-\delta T)} - 2G^{(T)}}{\delta T^2}$$

where C_p is the specific heat capacity, T is the temperature

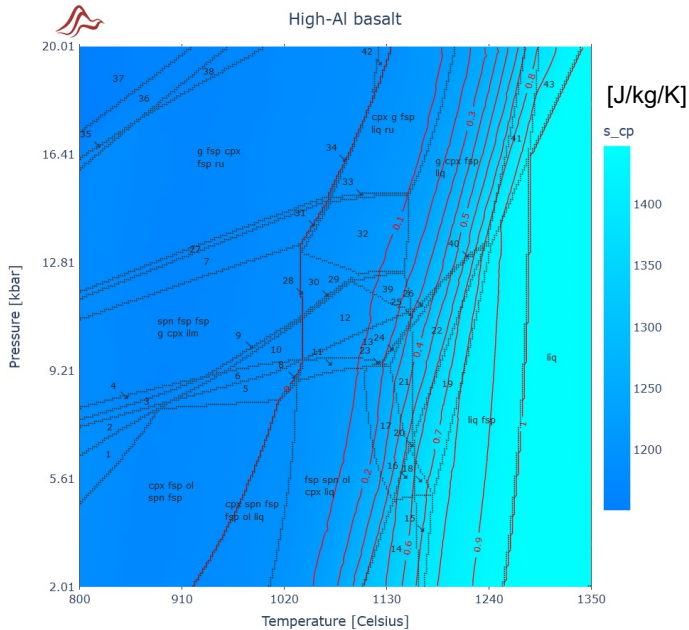
- Can be computed using

1. phase aggregate (G_0) **Without latent heat**
2. phase assemblage (G_{system}) **With latent heat**

- Note that when **G_system** is selected, solver must be set to legacy
higher accuracy at suprasolidus conditions is needed to compute Gibbs second derivative

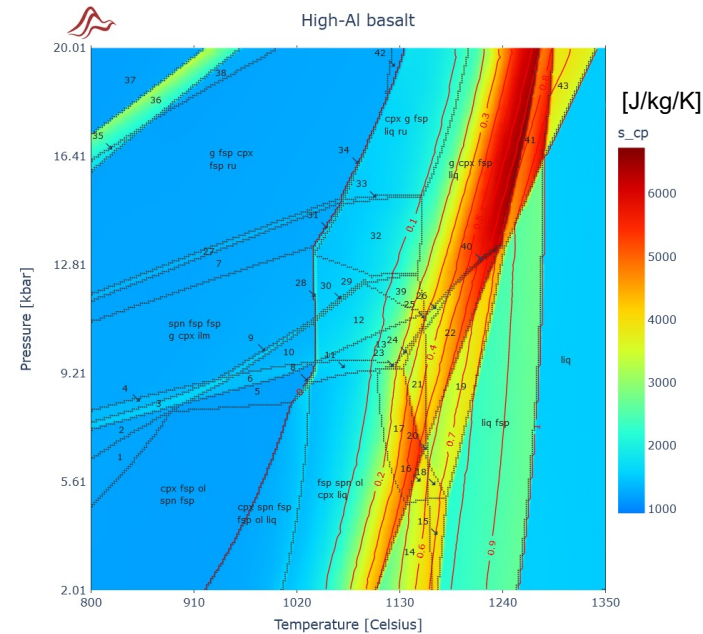
Specific heat capacity: what difference does it make?

G_0 differentiation
(Cp without latent heat of reaction)



- Cheap to compute
- Ignores latent heat

G_{system} differentiation
(Cp with latent heat of reaction)



- Slightly more expensive to compute
- Correct heat budget

MAGEMinApp: bulk rock composition

Bulk-rock composition

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous

oxide	mol_fraction
SiO ₂	38.494
Al ₂ O ₃	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K ₂ O	0.01
Na ₂ O	0.25
TiO ₂	0.1
O	0.096
Cr ₂ O ₃	0.109
H ₂ O	0

Buffer multiplier 0

ig

- Load custom bulk rock composition(s)
- Select predefined bulk rock composition for testing
- Bulk rock composition displayed in mol%
note that only the oxides taken into account in the database are displayed

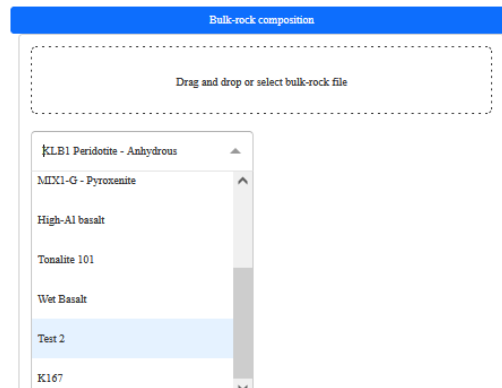
MAGEMinApp: bulk rock input file format

Example of input file provided in:

https://github.com/ComputationalThermodynamics/Resources/GG2024/bulkrock_composition_input_files/

```
# HEADER
title; comments; db; sysUnit; oxide; frac; frac2
# BULK-ROCK COMPOSITION
Test 2; Moo-et-al., 2000; ig; mol; [SiO2, Al2O3, CaO, MgO, FeO, K2O, Na2O, TiO2, O, Cr2O3, H2O]; [48.97, 11.76, 13.87, 4.21, 8.97, 1.66, 10.66, 1.36, 1.66, 0.01, 5.0];
K167; Moo-et-al., 2000; ig; mol; [SiO2, Al2O3, CaO, MgO, FeO, K2O, Na2O, TiO2, O, Cr2O3, H2O]; [50.3162, 8.0778, 11.6616, 11.3262, 9.1893, 0.4312, 2.4051, 2.271, 4.312, 0.0096, 0.0];
Test 3; Coin & Kwak, 1984; mb; wt; [SiO2, Al2O3, CaO, MgO, Fe2O3, K2O, Na2O, TiO2, FeO, H2O]; [55.12, 12.76, 4.32, 5.21, 2.45, 1.66, 10.66, 1.36, 1.66, 2.0];
Simon; Shorn-Simon, 2023; mb; mol; [SiO2, Al2O3, CaO, MgO, FeO, K2O, Na2O, TiO2, O, H2O]; [61.8, 8.57, 9.26, 5.05, 7.27, 0.21, 3.43, 1.25, 0.73, 2.85];
```

- **Note that either [..., Fe2O3, FeO,...] or [..., FeO, O, ...] must be provided for the database accounting for O**



- **Once loaded, the bulk-rock composition appear in the dropdown menu**
Note that the bulk are attached to a database, this is why only Test 2 and K167 appear in the dropdown menu



https://computationalthermodynamics.github.io/MAGEMin_C.jl/dev/

MAGEMin Toolset

Phase equilibrium modelling

A set of tools to compute phase equilibria for Earth-like compositions



Getting Started

MAGEMin_C.jl API

View on GitHub



MAGEMinApp

Effortlessly compute phase diagrams and Pressure-Temperature paths



MAGEMin_C

Set of function to freely compute phase equilibrium in Julia



MAGEMin

C program backend of MAGEMin

MAGEMinApp.jl

Introduction

Installation

Tutorials - Bulk input file

Tutorials - Phase diagrams

Tutorials - P-T-X paths

Tutorials - Isentropic paths

Citation

Interface

MAGEMin_C.jl

Introduction

Installation

Tutorials - Output

Tutorials - Step by step

Tutorials - Quick examples

API



MAGEMinApp: bulk rock input example

https://computationalthermodynamics.github.io/MAGEMin_C.jl/dev/MAGEMinApp/bulk_rock

this is a commented line

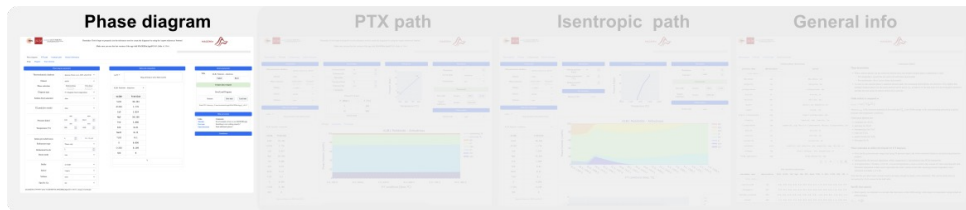
title; comments; db; sysUnit; oxide; frac; frac2

Test 2; Moo et al., 2000; ig; mol; [SiO₂, Al₂O₃, CaO, MgO, FeO, K₂O, Na₂O, TiO₂, O, Cr₂O₃, H₂O]; [48.97, 11.76, 13.87, 4.21, 8.97, 1.66, 10.66, 1.36, 1.66, 0.01, 0.0]; [48.97, 11.76, 13.87, 4.21, 8.97, 1.66, 10.66, 1.36, 1.66, 0.01, 5.0]

Test 3; Coin & Kwak, 1984; mb; wt; [SiO₂, Al₂O₃, CaO, MgO, Fe₂O₃, K₂O, Na₂O, TiO₂, FeO, H₂O]; [55.12, 12.76, 4.32, 5.21, 2.45, 1.66, 10.66, 1.36, 1.66, 2.0];

MAGEMinApp: phase diagram exercises

https://computationalthermodynamics.github.io/MAGEMin_C.jl/dev/MAGEMinApp/PD_tutorials



Phase diagrams tutorials (MAGEMinApp v0.8.6)

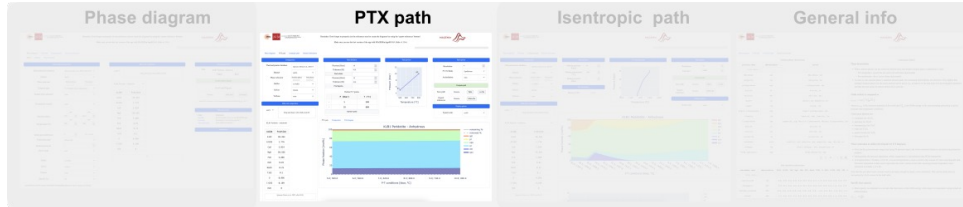
Here we provide a set of tutorials to generate various kind of phase diagrams, compute trace-element partitioning and Zr saturation, post-process the results, display various fields, reaction lines and iso-contours, and, export data and save the diagrams as [svg](#) graphic vector files.

Info

- [1. First phase diagram](#)
- [2. Reaction lines and isopleths](#)
- [3. Displayed field and colormap options](#)
- [4. Export figures](#)
- [5. Deactivate solution and pure phases](#)
- [6. Buffers](#)
- [7. Latent heat of reaction](#)
- [8. Trace element modelling](#)
- [9. Solidus H₂O saturated phase diagram](#)
- [10. TX fixed pressure diagram](#)
- [11. PTX diagram](#)
- [12. TT poly-metamorphic diagram](#)
- [13. LaMEM density diagram](#)

MAGEMinApp: P-T-X paths exercices

https://computationalthermodynamics.github.io/MAGEMin_C.jl/dev/MAGEMinApp/PTX_tutorials



P-T-X paths tutorials (MAGEMinApp v1.2.1)

Here we provide a set of tutorials to generate various kind of pressure-temperature-composition paths: including batch melting, fractional melting and fractional crystallization.

E.1. Quick start - first P-T-X path

For the first P-T-X path, simply launch [MAGEMinApp](#) and navigate to the [PTX path](#) tab. Change default setting for the thermodynamic database ([Igneous](#)) (Green et al., 2025 after Holland et al., 2018) and default bulk-rock composition ([KLB1 Peridotite anhydrous](#)). In the [Path definition](#) panel click on [Find solidus](#) and [Find liquidus](#) and define the P-T points accordingly: